Comprehensive Report on Peak Hour Traffic Analysis

Component 4: Data Visualisation & Analysis

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1. Introduction & Objective

This report identifies peak traffic hours at four road junctions using hourly vehicle-count data collected from 1 November 2015 to 30 June 2017, integrated with historical weather data and event/holiday information for Bengaluru. The objective is to determine when congestion is highest at each junction, how these peak periods vary by day of the week and by season, and how external factors such as rainfall and special events influence traffic volume. These findings are intended to support traffic management decisions such as signal timing adjustments and congestion mitigation planning.

Dataset Summary

- 48,120 hourly records across 4 junctions (Junctions 1-3: full range; Junction 4: Jan-Jun 2017 only)
- Enriched with hourly weather (temperature, humidity, precipitation, rain, wind speed)
- Enriched with public holiday flags and RCB/IPL home-match event flags for Bengaluru

2. Methodology

Congestion was measured using average vehicle count per hour, since direct speed or delay measurements were not available in the source data. For each junction, the mean vehicle count was computed for every hour of the day (0-23), and the hours with the highest mean values were treated as that junction's peak hours. Standard deviation was reviewed alongside the mean to confirm that identified peaks were consistent patterns rather than one-off spikes. The same aggregation was repeated separately by day-of-week (to compare weekdays and weekends) and by month (to capture seasonal variation), and cross-tabulated against weather and event flags to assess external influence.

3. Peak Hour Findings

Junctions 1, 2, and 3 all show a classic evening rush-hour pattern, peaking between 19:00 and 21:00. Junction 4 breaks this pattern entirely, peaking around midday (12:00-15:00) instead - a distinct behaviour worth flagging separately in any city-wide signal-timing plan, since a one-size-fits-all evening-rush intervention would miss Junction 4's actual congestion window.

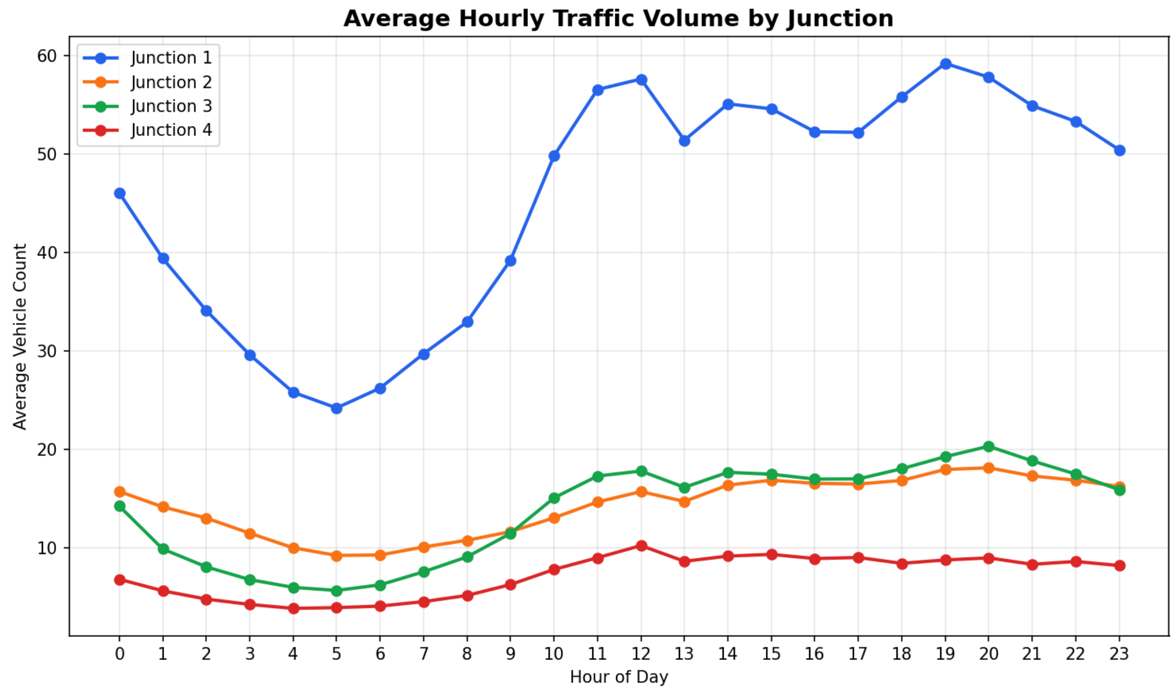

Figure 1: Average hourly traffic volume by junction. Junctions 1-3 peak in the evening; Junction 4 peaks at midday.

Peak hours by junction

Junction	Top 3 Peak Hours	Pattern
Junction 1	19:00, 20:00, 12:00	Evening rush + midday secondary peak
Junction 2	20:00, 19:00, 21:00	Evening rush
Junction 3	20:00, 19:00, 21:00	Evening rush
Junction 4	12:00, 15:00, 14:00	Midday peak - deviates from typical rush hour

The radial view below presents the same 24-hour pattern differently, making the shape of each junction's daily traffic rhythm easier to compare at a glance. Junction 1 clearly dominates in overall volume, with a wide plateau spanning 10:00 through 21:00 rather than a single sharp spike.

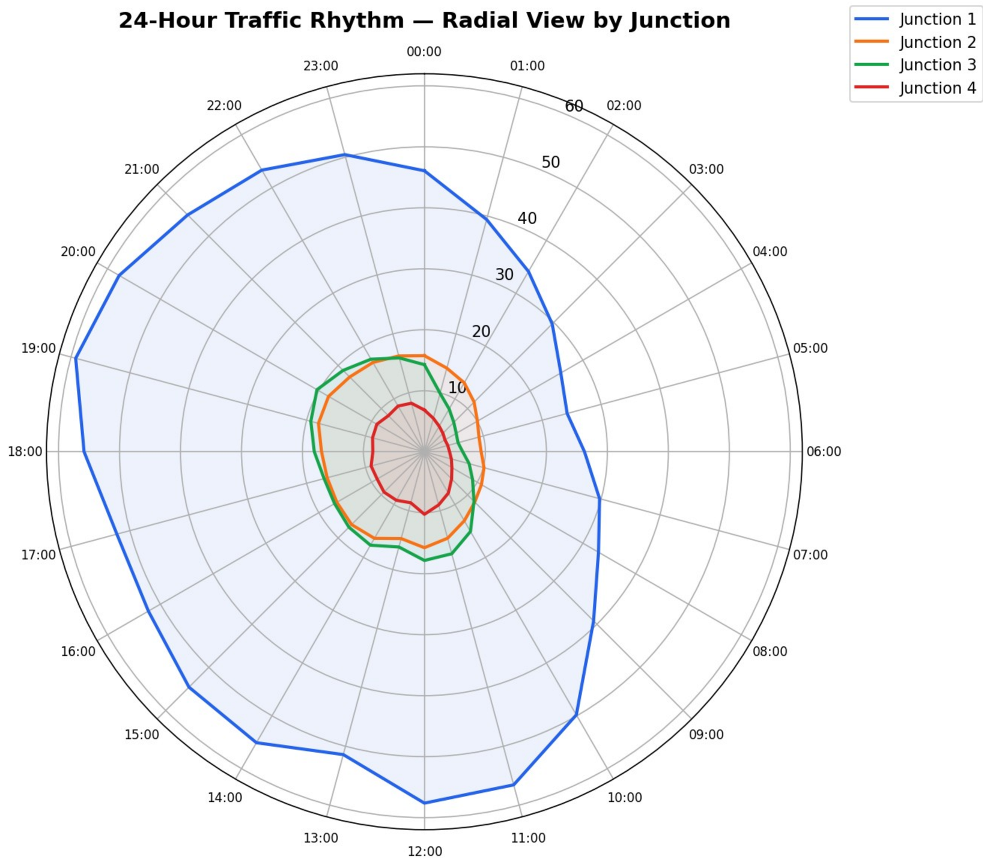

Figure 2: 24-hour traffic rhythm by junction, radial view.

4. Day-of-Week Patterns

The hour x day-of-week heatmaps below confirm that weekday congestion is markedly higher than weekend congestion across Junctions 1, 2, and 4, with the darkest cells concentrated on Monday-Friday during evening hours. Junction 3 shows an unusual Saturday-night spike around 20:00-21:00, distinct from its otherwise low weekend baseline - potentially linked to weekend leisure or nightlife traffic near that junction.

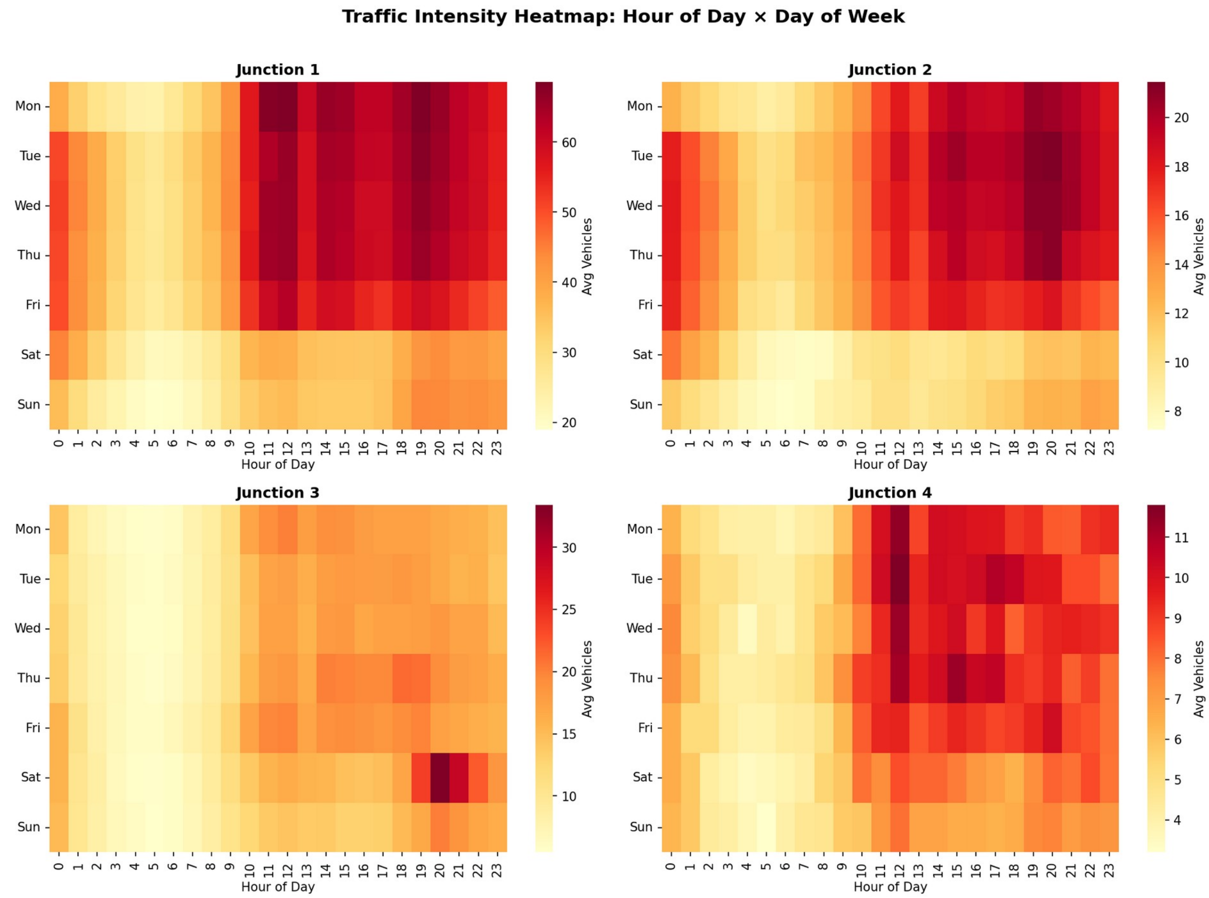

Figure 3: Traffic intensity heatmap (hour of day x day of week) per junction.

Aggregating all junctions together, weekday traffic peaks at 19:00 (average ~32.7 vehicles) while weekend traffic peaks slightly later, at 20:00 (average ~25.9 vehicles) - both lower in volume and shifted about an hour later than the weekday peak, consistent with later, more leisure-driven travel patterns on weekends.

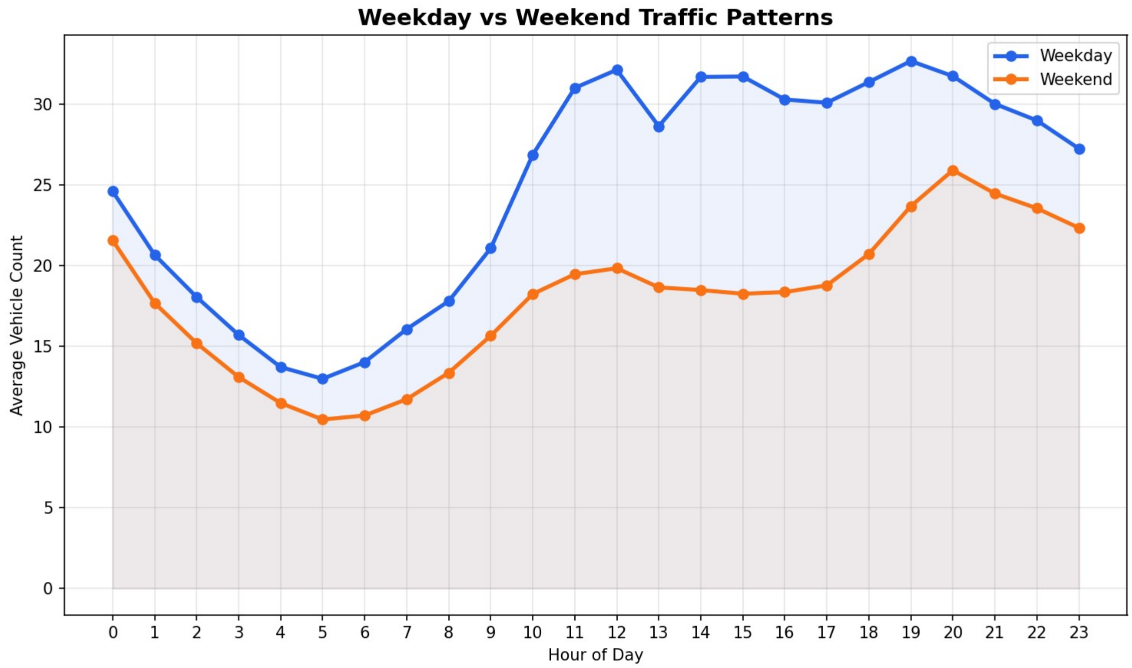

Figure 4: Weekday vs weekend hourly traffic comparison (all junctions combined).

5. Seasonal Patterns

Junction 1, by far the highest-volume junction, shows clear seasonal swings - traffic rises through the first half of the year to a peak around June, dips sharply in July, then partially recovers toward October. October, June, and September were the three highest-traffic months on average across the dataset. Junctions 2-4 show comparatively flatter, more stable monthly patterns, suggesting seasonal effects are strongest at the busiest junction and less pronounced at lower-volume ones.

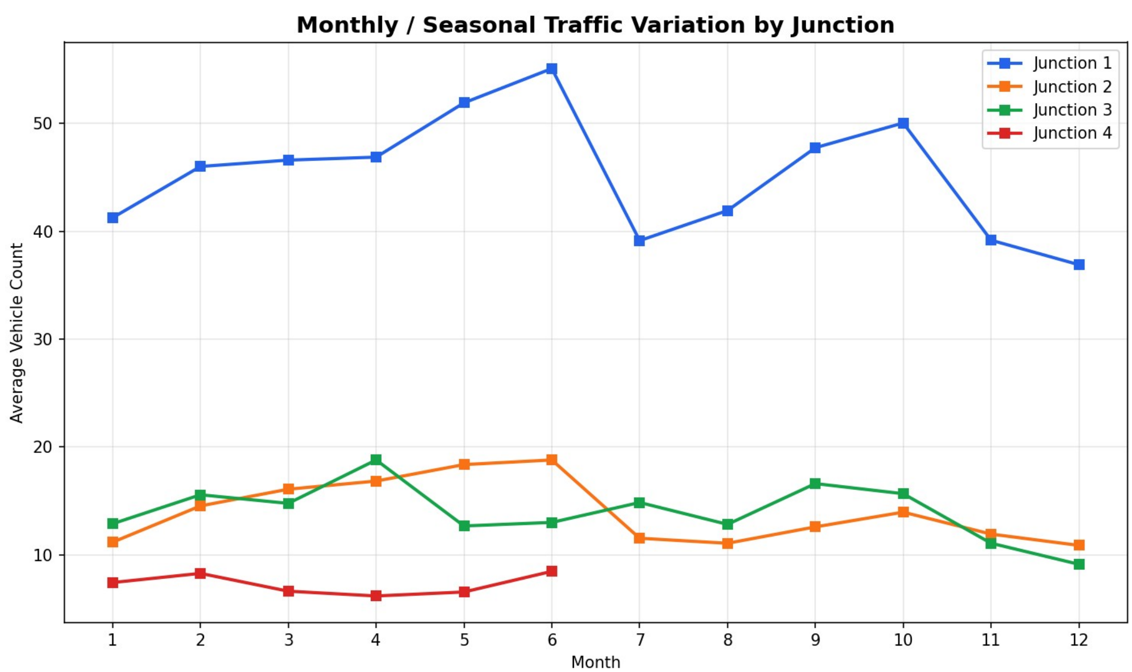

Figure 5: Monthly / seasonal traffic variation by junction.

6. External Influencing Factors

6.1 Rainfall

Comparing peak-hour traffic on dry versus rainy hours shows only a mild difference - average vehicle counts of 24.96 on dry hours versus 26.18 on rainy hours. Rain appears to slightly increase peak-hour congestion (likely due to slower vehicle speeds and more cautious driving, consistent with the traffic-and-pricing relationship discussed in the Component 2 report), but the effect is not large in this dataset.

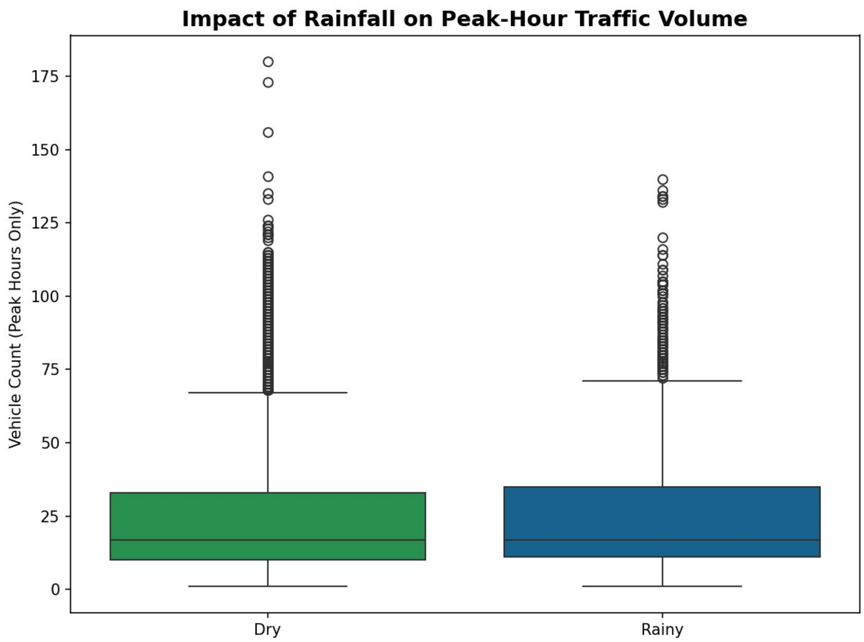

Figure 6: Distribution of peak-hour vehicle counts on dry vs rainy hours.

6.2 Holidays and Special Events

Contrary to the intuitive assumption that holidays and event days would increase congestion, peak-hour traffic was actually slightly lower on holidays (average 24.33) and special event days (average 22.52) compared to normal days (average 25.26). This is a genuinely counter-intuitive finding worth highlighting: it suggests that while events such as IPL matches concentrate traffic locally around the venue at specific times, they reduce general commuter traffic citywide during the same hours, since regular office and school commuting is absent. This distinction matters for traffic planning - event-day management should focus on venue-adjacent congestion rather than assuming city-wide peak-hour traffic will rise uniformly.

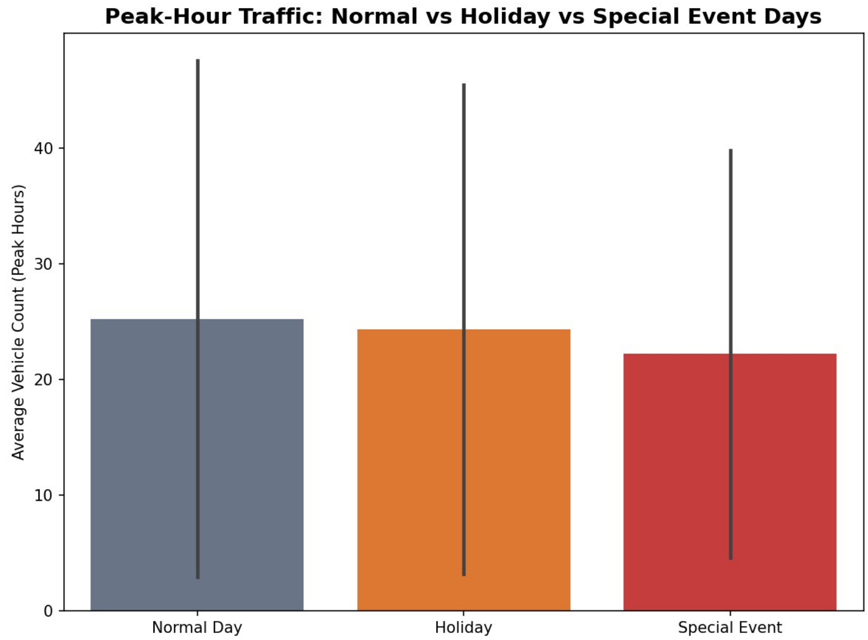

Figure 7: Peak-hour traffic on normal days vs holidays vs special event days.

7. Recommendations

- Signal timing at Junction 4 should be optimised for a midday (12:00-15:00) peak, not the evening rush window used elsewhere - applying a uniform city-wide schedule would under-serve this junction during its actual congestion period.
- Junctions 1-3 should continue to prioritise evening-rush interventions (19:00-21:00), where congestion is most consistent and severe.
- Given the Saturday-night spike unique to Junction 3, targeted weekend-evening measures (e.g. temporary signal adjustments) may be more effective there than blanket weekend timing changes.
- Since rainfall only mildly increases peak-hour congestion, weather-triggered traffic interventions can be lower priority compared to time-of-day and day-of-week factors.
- Event-day traffic management should be venue-focused (localised around Chinnaswamy Stadium during matches) rather than assuming a city-wide surge, since overall peak-hour volumes are actually slightly lower on those days.
- The reduced general commuter traffic observed on holidays presents a practical opportunity: scheduling planned road maintenance or infrastructure work on public holidays would cause comparatively less disruption.